6.2.1 Filling rectangular tiles using Tetrominoes

Situation 2

Can you form the rectangles of same area using the five tetrominoes only once? No, observe the following formations (*Fig 6.9 – 6.12*) where we have arranged tetrominoes edge to edge and tried to fill in rectangles. The boxes marked **X** are not filled by a tetromino. Hence the rectangle is incomplete. Also, some squares protrude outside the boundary of the rectangle.

The five tetrominoes together consist of 20 squares. Using 20 squares we can form the rectangles of size 1 x 20, 20 x 1, 2 x 10, 10 x 2, 4 x 5, and 5 x 4.

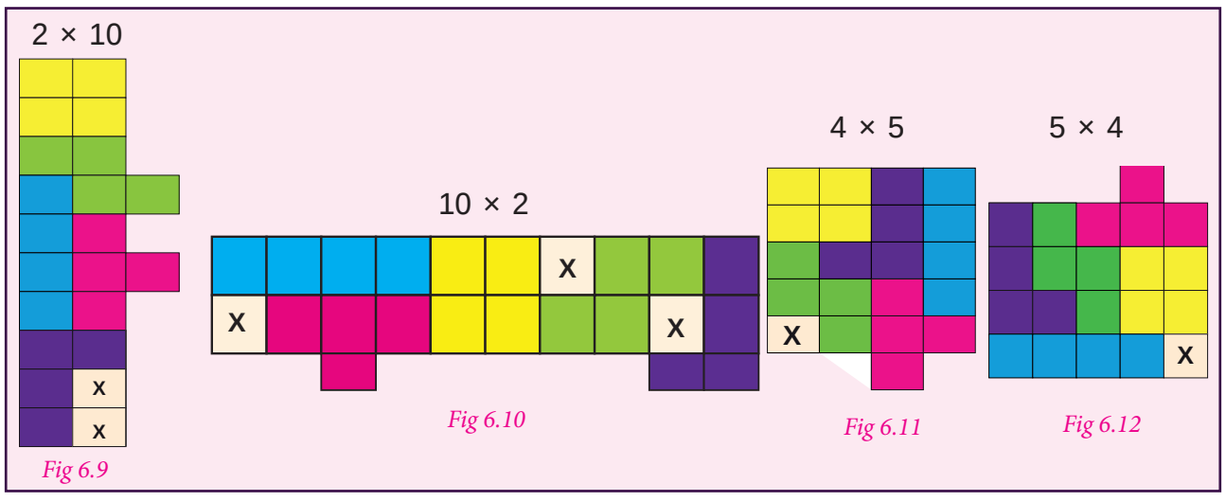


Fig 6.9 *Fig 6.10* *Fig 6.11* *Fig 6.12*

Why we are not able to form rectangular shapes using these tetrominoes only ones?. To know the reason, take the five tetrominoes in the form shown below.

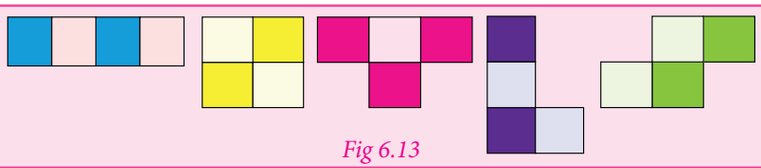


Fig 6.13

From among the five tetrominoes, in four of them, the number of shaded and unshaded squares are equal. But, in one tetromino  , the shaded and unshaded squares are not equal. Hence, it is impossible to fit the tetrominoes in the rectangle using it only once.

However, If we use all the five tetrominoes twice, rectangles of sizes 5 x 8 , 4 x 10 and so on. It can be completely filled in, as shown in Fig. 6.14 and Fig.6.15.

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5 × 8

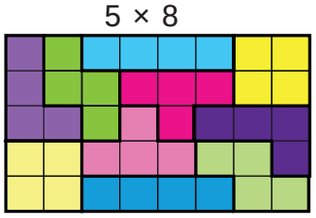


Fig 6.14

4 × 10

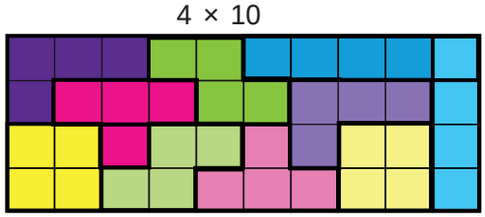


Fig 6.15

This concept will be useful in many places in real life situation like tiling a floor, packing things in a box and so on.

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